A Literature-Implied Partial Answer to Fremlin's Problem DU

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Status

This is a literature-implied partial answer to Fremlin's Problem DU as recorded in Aviles–Plebanek–Rodriguez, arXiv:1001.4896 [1]. The supporting result is due to Dodos–Kanellopoulos, arXiv:0805.2031 [2]. The identification is agent-made: the supporting paper studies Fremlin's filling families over the Cantor set, but the source paper still records the arbitrary uncountable-set version as open.

Source Problem

The source paper defines an ε-filling family on a set S to be a hereditary family $\mathcal{F} \subset [S]^{<\omega}$ such that for every finite $H \subset S$ there is $F \in \mathcal{F}$ with

$$F \subset H, \qquad |F| \geq \varepsilon |H|.$$

It calls $\mathcal{F}$ compact when $\{1_F : F \in \mathcal{F}\}$ is compact in 2^S, and records Fremlin's Problem DU:

> Are there compact ε-filling families on uncountable sets?

Supporting Theorem

Dodos–Kanellopoulos prove that if $\mathcal{F}$ is an ε-filling family over the Cantor set $\mathcal{C}$ and $\mathcal{F}$ is C-measurable in $[\mathcal{C}]^{<\omega}$, then for every perfect $P \subset \mathcal{C}$ there is a perfect $Q \subset P$ such that

$$[Q]^{<\omega} \subset \mathcal{F}.$$

In particular, they note that every C-measurable ε-filling family over $\mathcal{C}$ is not compact.

Implication

There is no compact ε-filling family on any set S with $|S| \geq \mathfrak{c}$.

Indeed, suppose $\mathcal{F} \subset [S]^{<\omega}$ is compact and ε-filling. Choose $T \subset S$ with $|T| = \mathfrak{c}$. The restriction

$$\mathcal{F}_T = \{F \in \mathcal{F} : F \subset T\}$$

is compact in 2^T, hereditary, and ε-filling over T. Transport $\mathcal{F}_T$ by a bijection between T and the Cantor set $\mathcal{C}$. This gives a compact ε-filling family $\mathcal{G} \subset [\mathcal{C}]^{<\omega}$.

Compactness in $2^{\mathcal{C}}$ makes $\mathcal{G}$ C-measurable in $[\mathcal{C}]^{<\omega}$. By Dodos–Kanellopoulos, there is a perfect infinite $Q \subset \mathcal{C}$ such that $[Q]^{<\omega} \subset \mathcal{G}$. But then the net of finite subsets of Q, directed by inclusion, converges pointwise in $2^{\mathcal{C}}$ to the infinite set Q. Since $\mathcal{G}$ is compact, hence closed in $2^{\mathcal{C}}$, it would contain Q, a contradiction to $\mathcal{G} \subset [\mathcal{C}]^{<\omega}$.

Scope

This gives a negative answer to Problem DU for all sets of cardinality at least $\mathfrak{c}$. In particular, under the Continuum Hypothesis, every uncountable set has cardinality at least $\mathfrak{c}$, so Problem DU has a negative answer under CH.

The arbitrary ZFC problem remains open for possible uncountable sets of cardinality strictly below $\mathfrak{c}$. This packet therefore belongs in the literature-implied partial-answer bucket, not in full solutions.

References

References

[1] A. Aviles, G. Plebanek, and J. Rodriguez, *The McShane integral in weakly compactly generated spaces*, arXiv:1001.4896.

[2] P. Dodos and V. Kanellopoulos, *On filling families of finite subsets of the Cantor set*, arXiv:0805.2031.